# Daily Combinatorics Drill #6

*Sheet by David Akinyele-Aje*

| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Pigeonhole Principle (basic & strong) · Worst-Case Thinking · Double Counting · The Handshake Lemma.*

## Section A Rapid Recognition ($\leq$15 s each)

**1.** Socks to guarantee a matching pair?

Ans: 3

**2.** Draws to guarantee a black pair?

Ans: 12

**3.** 13 people, two share a birth month?

Ans: True

**4.** Guarantee two share a weekday of birth?

Ans: 8

**5.** Integers to guarantee two of the same parity?

Ans: 3

**6.** Handshake counts total 18: how many handshakes?

Ans: 9

**7.** Row totals sum = column totals sum?

Ans: True

**8.** Guarantee two integers differing by a multiple of 5?

Ans: 6

**9.** Sweets to guarantee 3 of one flavour (4 flavours)?

Ans: $4 \cdot 2 + 1 = 9$

**10.** Any 4 integers: two with even difference?

Ans: True

## Section B Manipulation Drills ($\leq$50 s each)

**1.** 12 round-table seats: smallest number forcing an adjacent pair? *(after TMUA 2019 P2 Q9)*

Ans: 7

**2.** Guarantee two integers with difference divisible by 7?

Ans: 8

**3.** Two of any 5 integers differ by a multiple of 4; five integers avoiding multiples-of-5 differences.

Ans: Pigeonhole on 4 residues; construction: $1, 2, 3, 4, 5$

**4.** Six from $\{1, \ldots, 10\}$: two summing to 11? *(after TMUA 2022 P2 Q8)*

Ans: Yes; the smallest forcing size is 6

**5.** Round-robin with 8 teams: matches and per-team games?

Ans: $\binom{8}{2} = 28$ matches; each team plays 7

**6.** Friendship counts total 84: how many friendships?

Ans: 42

**7.** Four from $\{1, \ldots, 6\}$: two consecutive?

Ans: Forced: largest safe choice has 3 elements, e.g. $\{1, 3, 5\}$

**8.** Two party guests with equal handshake counts: why can't 0 and $n-1$ coexist?

Ans: Someone shaking all $n-1$ hands has shaken the non-shaker's hand — contradiction.

**9.** 25 sweets, 7 children: someone gets at least 4.

Ans: At least $\lceil 25/7 \rceil = 4$

**10.** 5 committees of 4; everyone on exactly 2: how many members?

Ans: $\frac{5 \cdot 4}{2} = 10$

## Section C Substitution & Structure ($\leq$90 s each)

**1.** 10 red, 10 green, 10 blue: guarantee 5 of some colour; guarantee 5 red?

Ans: Some colour: $3 \cdot 4 + 1 = 13$. Five red: $20 + 4 + 1 = 25$.

**2.** Seven from $\{1, \ldots, 12\}$: two sum to 13; six don't force.

Ans: Forced at 7; construction for 6: $\{1, 2, 3, 4, 5, 6\}$

**3.** Guarantee two integers with the same last digit?

Ans: 11

**4.** Mutilated chessboard: 31 dominoes?

Ans: Impossible.

**5.** Group size to guarantee two share both birth month and weekday?

Ans: $12 \cdot 7 + 1 = 85$

**6.** 20 clubs $\times$ 6 students; each student in 3 clubs: student count?

Ans: $\frac{20 \cdot 6}{3} = 40$

**7.** 1–9 around a circle: three adjacent summing $\geq 15$.

Ans: Some three adjacent numbers sum to at least 15.

**8.** Five from $\{1, \ldots, 8\}$: two differing by 4.

Ans: Forced.

## Section D Challenge (SMC / BMO1 difficulty)

**1.** Five points in a unit square: two within $\frac{\sqrt{2}}{2}$.

Ans: Proof below.

**2.** Five lattice points: some midpoint is a lattice point.

Ans: Proof below.

**3.** The number of people with an odd handshake count is even.

Ans: Proof below.

**4.** 20 students, each solving exactly 2 of 5 problems: two share a problem-pair.

Ans: Proof below; some pair chosen by $\geq 2$ students (in fact $\geq \lceil 20/10 \rceil = 2$).

**5.** Six people: three mutual friends or three mutual strangers.

Ans: Proof below.